EXPERIMENT - 3

Date: 2082-08-02

TO DETERMINE THE THICKNESS OF GIVEN GLASS PLATE BY USING SPHEROMETER AND HENCE CALCULATE ITS DENSITY.

APPARATUS REQUIRED:

1. Spherometer
2. The test plate
3. Base plate
4. Physical Balance

THEORY:

The pitch of the screw is defined as the linear distance travelled by the screw in one complete rotation of the circular scale.

Least Count (L.C.) is the smallest value that the instrument can measure accurately. In Spherometer, it is the distance travelled by screw when the circular scale is rotated through its one circular scale division.

Hence, if P be the pitch of the screw and N be the number of the circular scale divisions, then the least count (L.C.) of the Spherometer is given by,

$$L.C. = \frac{\text{Pitch}}{\text{No. of circular scale division}}$$

or, $$L.C. = \frac{P}{N}$$

Let, t = thickness of the given test plate

A = area of the glass plate

Then, volume of the glass plate (V) = A × t

If m be the mass of the test plate, then its density is given by

$$\text{Density } (\rho) = \frac{\text{Mass}}{\text{Volume}} \quad \text{i.e. } \rho = \frac{m}{V}$$

OBSERVATIONS:

No. of circular scale divisions (N) = ...100......... divisions

Value of 10 smallest division of the main scale = ...10............ mm

∴ Value of 1 smallest division of the main scale = ...1........... mm

In 4 complete rotations, the circular scale moves through ...4..... divisions of the main scale.

In 1 complete rotations, the circular scale moves through ....1..... divisions of the main scale.

∴ Pitch (P) = ...1........... mm

Now, Least Count (L.C.) = $\frac{P}{N}$ = ..0.01....... mm

Mass of the given test plate, (m) = ...4.10....... g

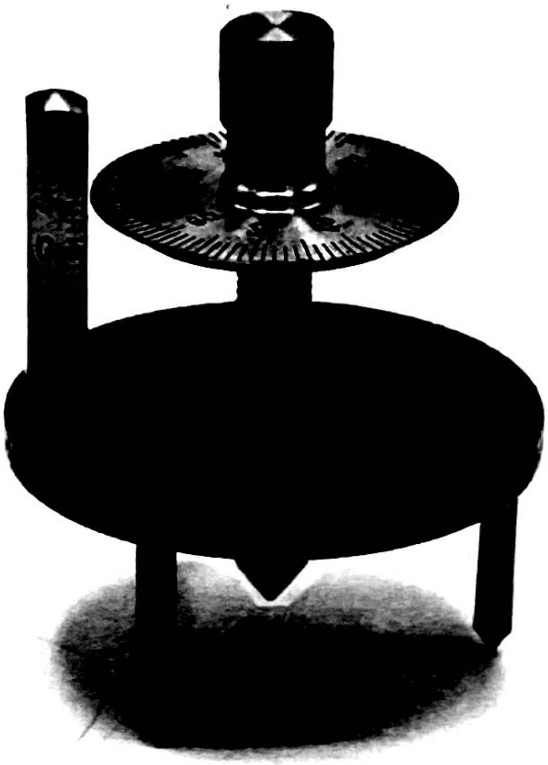
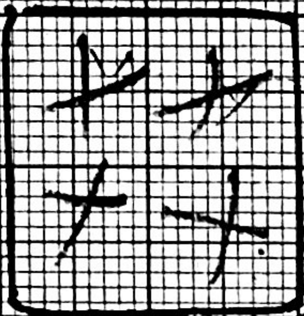
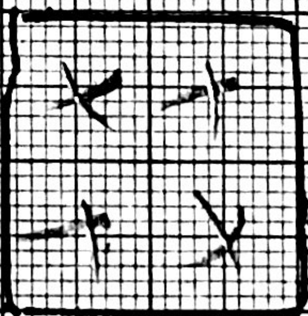


Fig. Spherometer



400
+400
800
400



4/2
400
800

$2 = \frac{400}{100} = 4$

Table for Thickness

No. of obs.	Initial circular scale reading on test plate (a)	No. of complete rotations made (x)	Value of complete rotations = x × P (A) mm	Final circular scale reading on base plate (b)	No. of additional circular scale divisions N = (a – b) for a > b, or = (100 + a – b) for a < b	Value of N = N × L.C. (B) mm	Total thickness = A + B mm	Mean thickness (t) mm
1	45	4	4	63	82	0.82	4.82	4.77
2	44	4	4	63	81	0.81	4.81	
3	45	4	4	63	82	0.82	4.82	
When the test plate is turned over								
1	43	4	4	71	72	0.72	4.72	
2	44	4	4	71	73	0.73	4.73	
3	44	4	4	71	73	0.73	4.73	

∴ Mean thickness (t) = 4.77 mm
= 0.47 cm

Value of area of 1 small square of graph paper i.e. L.C= 1 mm²

Table for area of glass plate

Face	No. of complete small square (a)	No.of incomplete square (b)	Total (a+b)	Area of plate (a+b) × L.C. mm²	Mean area A mm²
1	400	0	400	400	400
2	400	0	400	400	

From graph, the area of the test plate (A) = 400 mm²
= 4 cm²

CALCULATION:

Volume of the given glass plate (V) = Area (A) × Thickness (t)
= 4 x 0.47
= 1.88 cm³

Density of the test plate (ρ) = $\frac{m}{V}$
= 4.18/1.88
= 2.22 g/cm³

RESULT:

Hence the thickness, volume and density of the given glass plate are found to be 0.47 cm, V= 1.88 cm³ ρ=2.22 g/cm³.

PERCENTAGE ERROR:

Now, standard value of density (S.V.) = 2.5 g/cm³ (for glass)

$$\% \text{ Error} = \left|\frac{S.V. - O.V.}{S.V.}\right| \times 100\%$$

$$= \frac{2.5 - 2.22}{2.59} \times 100\%$$

$$= 10.81\ \%$$

CONCLUSION:

Hence from this experiment, we determined the value and density of the given glass with 10.81% error.

SOURCES OF ERROR:

1. Backlash error of the screw
2. The defect in graph paper
3. Carelessness in experiment
4. Blunt pencil
5. Irregular thickness of glass.

PRECAUTIONS:

1. The mass of the Test ~~tab~~ (glass) plate should be accurate.
2. The base plate should be cleaned.
3. Use magnifier to find out the division.
4. Observation should be taken at different point.

VIVA QUESTIONS:

1. Why the name Spherometer so called?
2. What is the principle of Spherometer?
3. What do you mean by radius of curvature of spherical surface?
4. Is paper insertion method for testing the touching position of the screw is correct?
5. What is meant by backlash error?
6. What shall be the effect on the least count of Spherometer if number of divisions on its circular scale be doubled?
7. What do you mean by the pitch of the Spherometer?
8. How can you make the Spherometer more sensitive?
9. Why zero error is not considered in case of Spherometer?

1. It is used to determine the radius of curvature of a spherical surface so it is spherometer

2. It works on the principle of screw gauge i.e the rotation of the circular scale is directly proportional to the linear distance moved by its circular scale.

3. The distance between pole and centre of curvature of spherical surface is called radius of curvature of spherical surface.

4. No because of the thickness of ordinary paper is greater than least count.

5. Backlash error is the slight movement that occurs when the direction of motion is reversed in screw.

6. If the no. of circular division is increased the accuracy increases since least count decreases.

7. The linear distance covered by the circular disc in one complete rotation along the main scale is called pitch of the spherometer.

8. We can make spherometer more sensitive by increasing the no. of circular division by which the least count decreases and accuracy increases.

9. Zero error is not considered in case of spherometer because the device is designed in such a way that is automatically accounts for its own initial reference point.